

CodeCoins

Technical Documentation — Architecture, Data Model & User Flows

Version 1.0 · June 2026 · Deployment: <https://codecoins-v3.vercel.app>

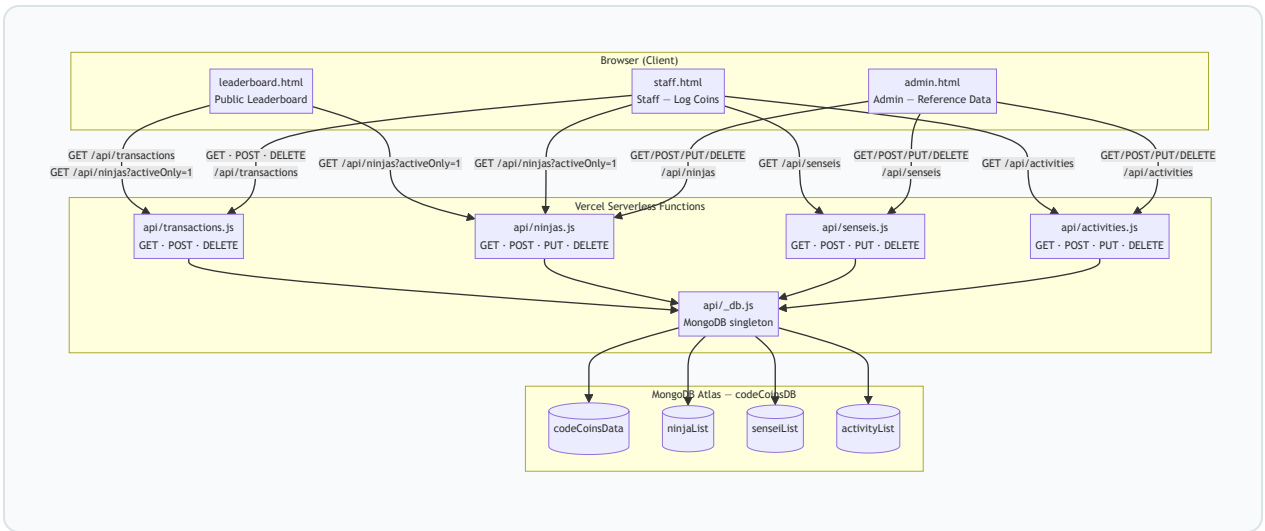
1. System Overview

CodeCoins is a gamification web application for a coding school. Senseis (instructors) award or redeem Code Coins to Ninjas (students) for completing activities. A live leaderboard shows each Ninja's running coin balance. Reference data (Ninjas, Senseis, Activities) is managed through a password-protected Admin page.

Layer	Technology
Frontend	Static HTML + Vanilla JavaScript
Backend	Vercel Serverless Functions (Node.js)
Database	MongoDB Atlas — <code>codeCoinsDB</code>
Hosting	Vercel — https://codecoins-v3.vercel.app

2. Architecture Diagram

The diagram below shows how the three frontend pages communicate with the Vercel Serverless Functions, which in turn connect to MongoDB Atlas through a shared connection singleton.



3. Data Model

ninjaList

Field	Type	Notes
<code>_id</code>	ObjectId	MongoDB auto-generated
<code>fullName</code>	String	e.g. "Amira Johnson" — used in Staff dropdown
<code>shortName</code>	String	Nickname used in transaction records
<code>status</code>	String	"Active" or "Inactive"

Legacy CSV field names: `Ninja Full Name`, `Ninja`, `Status`. The `api/ninjas.js` serialiser normalises both formats before returning data.

senseiList

Field	Type	Notes
<code>_id</code>	ObjectId	MongoDB auto-generated
<code>name</code>	String	Sensei display name

Legacy CSV field: `Sensei`.

activityList

Field	Type	Notes
<code>_id</code>	ObjectId	MongoDB auto-generated
<code>name</code>	String	Activity description
<code>coins</code>	Number	Default coin value for this activity

codeCoinsData

Field	Type	Notes
<code>_id</code>	ObjectId	MongoDB auto-generated
<code>date</code>	String	ISO date string of the transaction
<code>ninja</code>	String	Ninja short name
<code>sensei</code>	String	Sensei name
<code>activity</code>	String	Activity name
<code>pointsEarned</code>	Number	Coins awarded in this transaction
<code>pointsRedeemed</code>	Number	Coins redeemed in this transaction
<code>notes</code>	String	Optional free-text notes

Legacy CSV field names: `Ninja Nick Name`, `Points Earned`, `Points Redeemed`. The leaderboard reads both formats when aggregating balances.

4. API Reference

All endpoints are Vercel Serverless Functions under `/api/`. CORS is open (`*`) on all methods.

/api/transactions

Method	Params / Body	Description
<code>GET</code>	—	Returns all records from <code>codeCoinsData</code> , sorted newest first
<code>POST</code>	JSON body (transaction fields)	Inserts a new transaction record
<code>DELETE</code>	<code>?id=<ObjectId></code>	Deletes a single transaction

/api/ninjas

Method	Params / Body	Description
GET	?activeOnly=1 (optional)	Returns all ninjas; with flag, excludes documents where <code>status</code> / <code>Status</code> matches <code>/inactive/i</code> via MongoDB <code>\$nor</code>
POST	{ fullName, shortName, status }	Creates a new ninja
PUT	{ _id, fullName?, shortName?, status? }	Updates a ninja; writes both camelCase and legacy CSV field names
DELETE	?id=<ObjectId>	Deletes a ninja

/api/senseis

Method	Params / Body	Description
GET	—	Returns all senseis sorted by name
POST	{ name }	Creates a new sensei
PUT	{ _id, name }	Updates a sensei
DELETE	?id=<ObjectId>	Deletes a sensei

/api/activities

Method	Params / Body	Description
GET	—	Returns all activities sorted by name
POST	{ name, coins }	Creates a new activity
PUT	{ _id, name?, coins? }	Updates an activity
DELETE	?id=<ObjectId>	Deletes an activity

5. Page Descriptions

leaderboard.html — Public Leaderboard

- Dark animated background; no login required

- Fetches all transactions and all active ninjas in parallel on load
- Aggregates `pointsEarned` and `pointsRedeemed` per ninja client-side
- Coins balance = `pointsEarned - pointsRedeemed`; displays `"-"` when balance < 0
- Only ninjas present in `ninjaList` with `status = Active` appear on the board
- Sorted descending by Coins balance; ranked 1st, 2nd, 3rd with medal badges
- Live search filters the rendered table without re-fetching from the API
- Hamburger menu (top-right) navigates to Staff and Admin pages

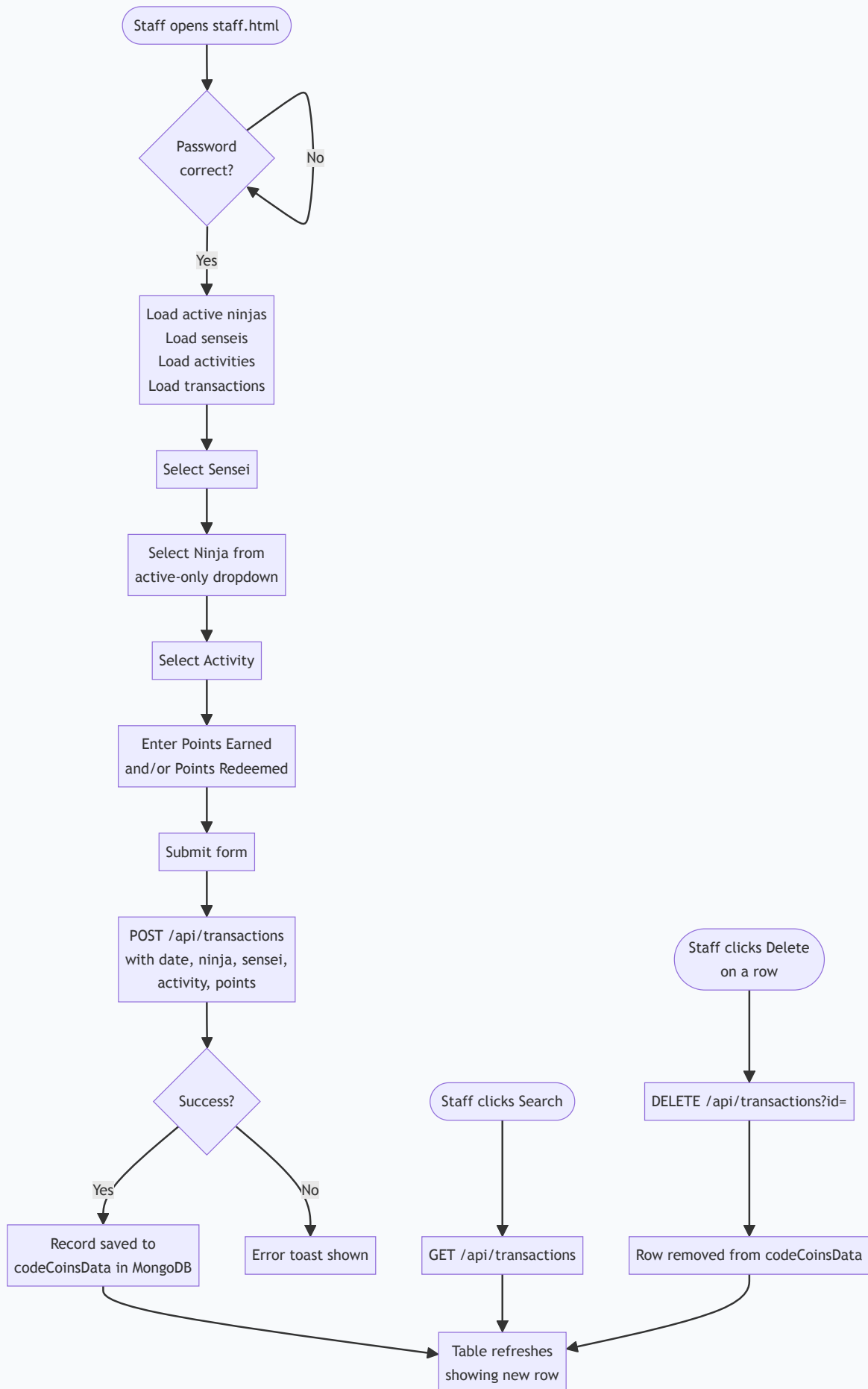
staff.html — Staff Coin Entry

- Password-protected lock screen on load
- Loads active ninjas, all senseis, and all activities from API on init
- Ninja dropdown shows only active ninjas (full name, sorted A–Z)
- Submitting the form POSTs a transaction record to `/api/transactions`
- Transaction history table supports search, date range filter, and "Search" refresh button
- Individual transaction rows can be deleted
- Hamburger menu navigates to Leaderboard and Admin pages

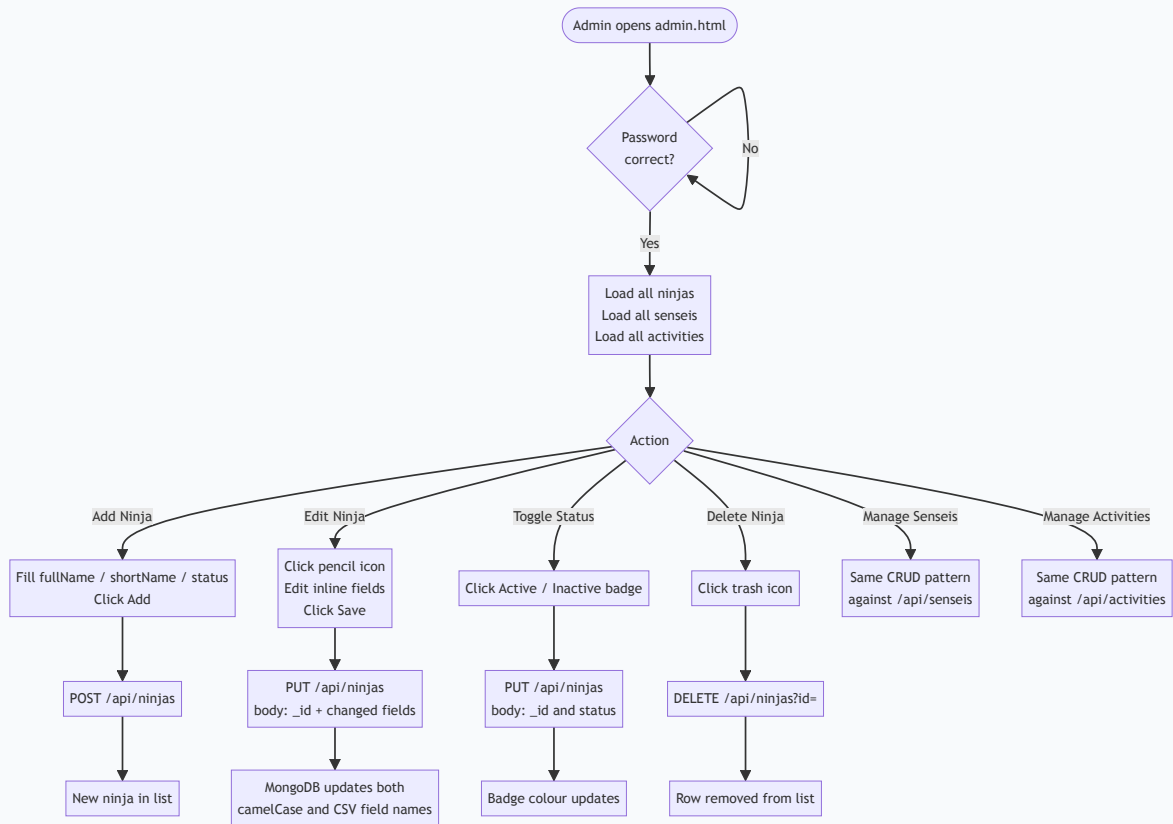
admin.html — Reference Data Management

- Password-protected lock screen on load
- Manages three reference collections: Ninjas, Senseis, Activities
- Each collection card supports Add (POST), inline Edit (PUT), and Delete
- Status toggle on each ninja row switches between Active/Inactive
- Hamburger menu navigates to Leaderboard and Staff pages

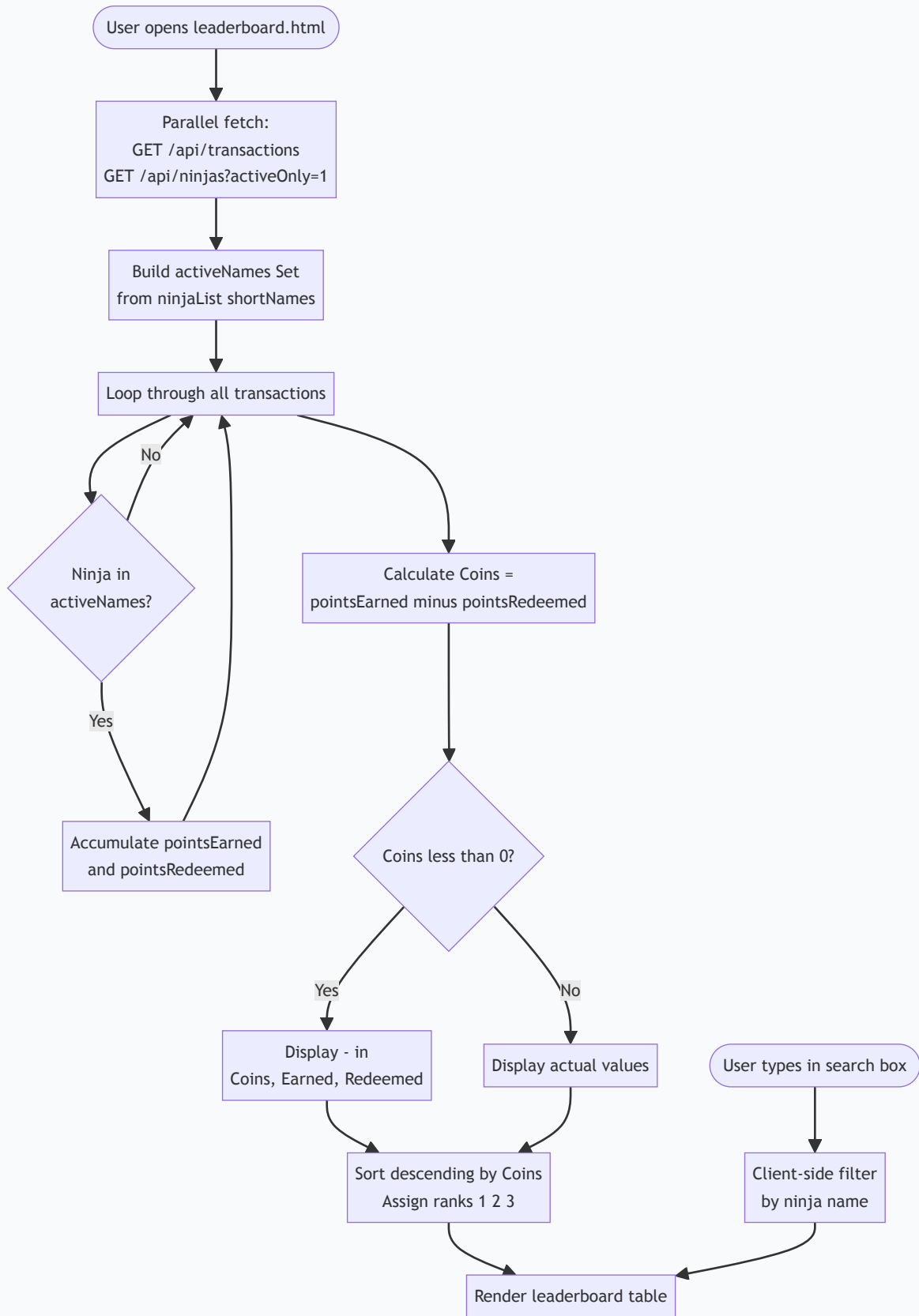
6. User Flow — Staff: Log Coins



7. User Flow — Admin: Manage Reference Data



8. User Flow — Leaderboard



9. Deployment

Item	Detail
Platform	Vercel
Production URL	https://codecoins-v3.vercel.app
Output directory	<code>.</code> (project root — all HTML files served as static)
Serverless runtime	Node.js (Vercel auto-detected from <code>api/</code> folder)
Environment variable	<code>MONGODB_URI</code> — MongoDB Atlas connection string (set in Vercel dashboard)
Deploy command	<code>npx vercel --prod --yes</code>

MongoDB Connection Singleton (`api/_db.js`)

The `_db.js` module keeps a single `MongoClient` instance alive across warm serverless invocations, avoiding a new TCP connection on every request:

```
const { MongoClient } = require('mongodb');
let client = null;

async function getDb() {
  if (!client) {
    client = new MongoClient(process.env.MONGODB_URI);
    await client.connect();
  }
  return client.db('codeCoinsDB');
}

module.exports = { getDb };
```

File Structure

```
CodeCoins/
├─ index.html # Landing / redirect page
├─ leaderboard.html # Public leaderboard
├─ staff.html # Staff coin-entry page
├─ admin.html # Admin reference-data page
├─ vercel.json # Vercel config (cleanUrls, outputDirectory)
├─ package.json
├─ Assets/ # Images and static assets
└─ api/
    └─ _db.js # MongoDB connection singleton
```

```
└─ transactions.js # /api/transactions → codeCoinsData
└─ ninjas.js # /api/ninjas → ninjaList
└─ senseis.js # /api/senseis → senseiList
└─ activities.js # /api/activities → activityList
```